PREDICTING ENERGY CONSUMPTION

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1. Introduction

Accurate forecasting of electricity demand is essential for efficient power system operation, grid stability, and energy resource planning. This project aims to predict hourly electricity consumption using historical load data from a transmission system operator. By applying machine learning techniques, the study explores how temporal patterns and historical dependencies can be leveraged to produce reliable short-term forecasts.

2. Dataset Description

The dataset contains approximately six years of hourly electricity consumption data.
Each record represents energy usage (in MWh) for a specific hourly interval.

Key Variable

- **Consumption (MWh)** — Target variable representing electricity demand

The time column was used as the primary index for time-series analysis.

➔ Some time is not rounded so I round it with floar.

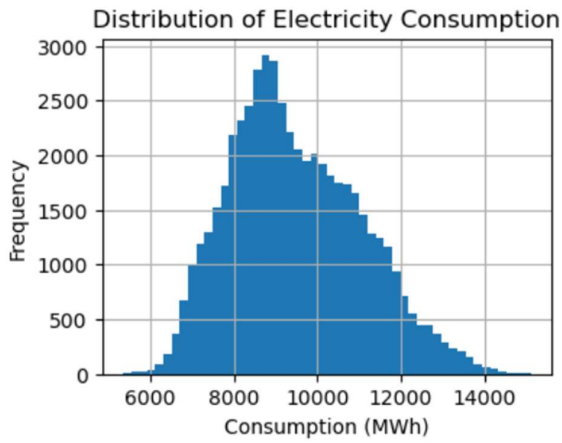
Eg. 10:05 rounded to 10:00

➔ Handled outliers.

➔ Checked Missing values but there are no missing values.

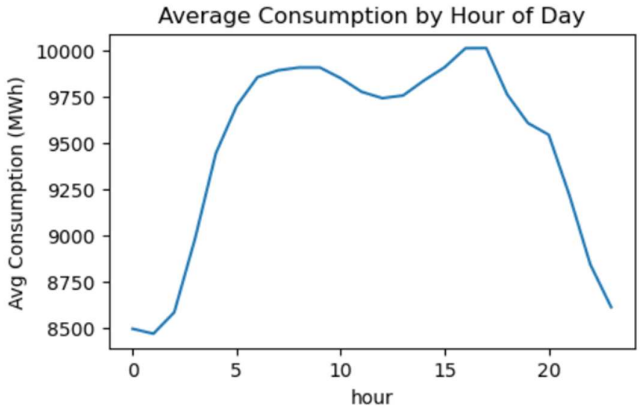
3. Exploratory Data Analysis (EDA)

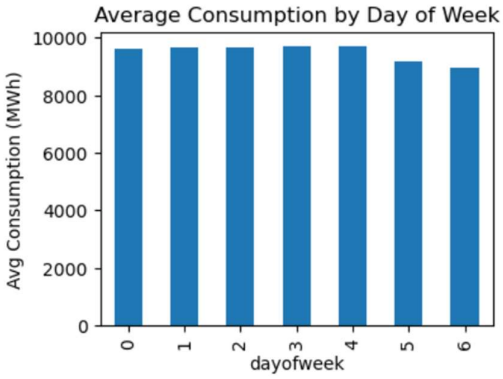
Several analyses were performed to understand the data:



Consumption is concentrated around a central range, with fewer extreme values, indicating generally stable demand with occasional peaks.

Average consumption is lowest during late night (around **8.4–8.6k MWh**) and gradually rises through the morning, peaking in the late afternoon (**≈10k MWh**), before declining again at night. This shows a clear daily demand cycle.





Average consumption remains fairly stable on weekdays (**≈9.6–9.8k MWh**), while it drops on weekends (**≈9.0–9.2k MWh**), indicating lower demand during non-working days.

4. Data Preprocessing

To ensure consistency and model readiness:

- Converted time column to datetime and set it as the index.
- Aligned timestamps to hourly intervals and merged duplicates.
- Handled missing values using time interpolation.
- Detected and capped outliers using the IQR method.
- Standardized the target column as **consumption**.

After preprocessing, the dataset contained a clean and continuous hourly time series.

5. Feature Engineering

To capture temporal patterns and dependencies, several features were created:

Temporal Features

- Hour of day
- Day of week
- Month
- Weekend indicator

Cyclical Encoding

Sin and cosine transformations were applied to represent cyclical time patterns smoothly.

Lag Features

- Previous hour consumption
- Previous day consumption
- Previous week consumption

Rolling Statistics

- 24-hour rolling mean
- 24-hour rolling standard deviation

These features allow models to capture seasonality, persistence, and short-term trends.

6. Model Selection

Three machine learning models were trained and compared:

1. Random Forest Regressor
2. XGBoost Regressor
3. Linear Regression

Hyperparameter tuning was performed using GridSearchCV for tree-based models.

JUSTIFICATION :-

Multiple models, including Random Forest, XGBoost, and Linear Regression, were evaluated for forecasting electricity consumption. Among them, **XGBoost was selected as the primary model** because it achieved the highest accuracy and effectively captured nonlinear relationships and seasonal patterns in the data.

Tree-based ensemble models are well-suited for this task as they handle temporal dependencies and feature interactions efficiently. Random Forest provided a strong baseline, while Linear Regression offered interpretability but showed lower performance during peak demand periods. Therefore, XGBoost was chosen as the most appropriate model for this forecasting problem.

7. Model Evaluation

Performance Metrics

Model	MAE	RMSE	R^2
Random Forest	70.31	96.70	0.9967
XGBoost	**62.47**	**84.83**	**0.9974**
Linear Regression	141.54	182.95	0.9881

INTERPRETATION :-

- **XGBoost achieved the best performance**, demonstrating superior ability to model complex temporal relationships.
- Random Forest provided a strong and stable baseline.
- Linear Regression showed higher errors, especially during peak demand periods.

8. Feature Importance Analysis

Across all models, the most influential predictors were:

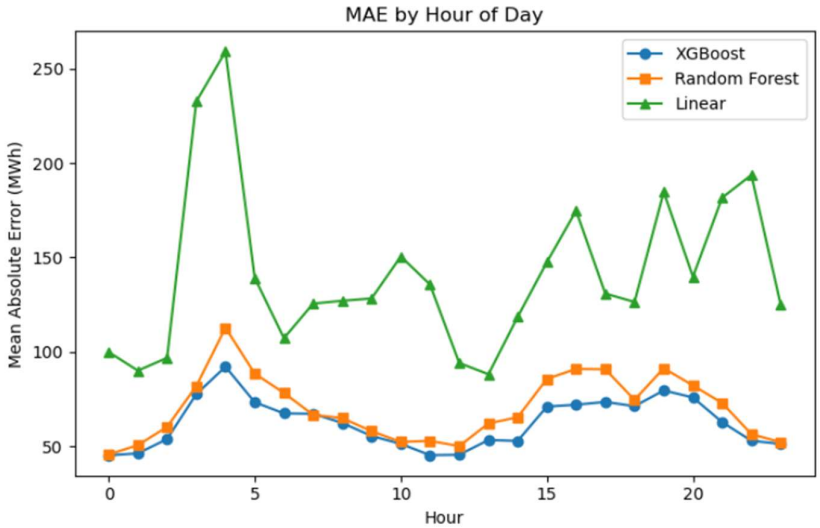
- Lagged consumption values
- Rolling statistics
- Cyclical time features

This confirms strong autocorrelation and seasonal patterns in electricity demand.

9. Error Analysis

Hourly error analysis showed:

- Slightly higher prediction error during peak demand hours
- Stable performance across most periods
- Tree-based models produced more consistent predictions than the linear model



10. Next Year Predictions

- **Approx hourly consumption next year:** Around **9,486 MWh per hour**, indicating expected demand to remain close to the historical average level.
- **Approx total next-year consumption:** Approximately **83.1 million MWh**, suggesting overall yearly demand is projected to stay consistent with past consumption patterns.

11. Key Insights

- Electricity demand shows a clear **daily cycle**, with higher usage during daytime and evening hours.
- **Weekday consumption is higher** than weekends, reflecting industrial and commercial activity.
- Demand is highly influenced by **recent past values**, indicating strong temporal dependency.
- Tree-based models (especially XGBoost) **predict demand more accurately**, particularly during peak periods.
- Overall consumption patterns are **stable over time**, with only occasional spikes.

12. Limitations

- No external factors (e.g., temperature or holidays) were included.
- Sudden abnormal demand spikes are harder to predict.
- Model relies solely on historical consumption patterns.

13. Conclusion

This study demonstrates that machine learning techniques can accurately forecast hourly electricity demand. Among the evaluated models, **XGBoost provided the highest predictive accuracy**, making it the most suitable approach for this dataset. The results highlight the importance of temporal feature engineering and confirm that energy demand exhibits strong persistence and seasonality. Such predictive models can play a crucial role in improving energy planning, optimizing resource allocation, and enhancing grid stability.